

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

20th March 2017

Subject ST4 – Pensions and Other Employee Benefits

Time allowed: Three Hours (14.45* – 18.00 Hours)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. *Please read the instructions inside the cover page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception.*
2. ** You have 15 minutes at the start of the examination in which you are required to read the questions. You are strongly encouraged to use this time for reading only, but notes may be made. You then have three hours to complete the paper.*
3. *You must not start writing your answers in the answer sheet until instructed to do so by the supervisor.*
4. *The answers are not expected to be any country or jurisdiction specific. However, if Examples/illustrations are required for any answer, the country or jurisdiction from which they are drawn should be mentioned.*
5. *Attempt all questions, beginning your answer to each question on a separate sheet.*
6. *Mark allocations are shown in brackets.*
7. *Please check if you have received complete Question paper and no page is missing. If so, kindly get a new set of Question paper from the Invigilator.*

AT THE END OF THE EXAMINATION

Please return your answer booklet and this question paper to the supervisor separately. You are not allowed to carry the question paper in any form with you.

Q. 1) The Government of a developed country wishes to reduce its citizens' dependence on state benefits.

i) List eight options available to the Government to encourage the private provision of pension payable on retirement or death. (4)

ii) State six methods which could be used to ensure that the system is not abused. (3)

The Government makes changes to the tax incentives for employers providing retirement benefits and, as a result, Company A is setting up a defined contribution pension scheme. The finance director of Company A has asked you to provide a table of contribution rates to use. He wants the contribution rate for each member to be fixed on joining the scheme. He has specified that the accumulated fund at retirement age of 60 should be sufficient to provide an annual income in retirement of 1% of final salary for each year of service and a spouse's pension of 50% of the member's pension. After retirement, the income should increase in line with price inflation. The benefits will be paid by the scheme and will not be insured.

iii) Describe briefly the method you would use to assess the contribution rates to be paid. (3)

iv) a) List the assumptions which you will have to make; and

b) Comment briefly on the factors you would take into account when choosing values for those assumptions. (6)

v) Set out the points which you would make to the finance director when presenting the table of contribution rates. (6)
[22]

Q. 2) i) Describe the features and purpose of the following types of valuations:

- Funding valuation
 - Discontinuance valuation
- (6)

ii) Describe the differences between a discounted cashflow funding valuation and a market value approach. (5)

iii) Outline the reasons why either a discounted cashflow approach or a market value approach might be used. (3)
[14]

Q. 3) A developing country is currently experiencing a sustained period of low inflation and low real investment returns.

i) Discuss the possible impact on the expectation of cost and future benefit design for the country's final salary pension plans. (8)

ii) Outline the possible impact on the members of defined contribution pension plans. (6)
[14]

Q. 4) A large company has hired an actuary to carry out the triennial valuation of its defined benefit pension scheme. The last valuation showed that there was surplus in the scheme.

- i) List the demographic and economic assumptions needed for a valuation of a defined benefit pension scheme. (4)

The Finance Director has asked the actuary to discuss the assumptions and rationale behind choosing the level of prudence.

- ii) Discuss the different types of bases used by the Actuaries and rationale behind choosing them. (4)

The Government of this developing country has demonetized the higher currency notes last year which resulted in a reduction in the interest rates during that year. However there was no significant impact on the overall GDP of the country and the interest rates later got restored to the same level before demonetization.

One of the trustees has said that the discount rate assumption does not appear to be prudent given that demonetization has lowered interest rates.

- iii) Discuss the considerations the actuary should make in using past data to set the discount rate and how he might allow for the fluctuation caused by last year's demonetization. (6)

The finance director is also concerned with the increasing level of premium being paid to insure the death in service benefits provided by the scheme.

These benefits are currently:

- A lump sum of ten times the member's basic salary.
- A non-increasing partner's pension equal to 80% of the member's prospective pension to normal retirement date(?).

- iv) Set out the points that the actuarial adviser to the company may include in a letter to the finance director explaining why the premiums may have increased. (3)

- v) Suggest possible ways in which the death in service insurance premiums could be reduced. (4)

[21]

Q. 5) A medium-sized company has been experiencing financial difficulties and has not been able to fund its defined benefit pension scheme for the last year. The security of benefits is a cause of concern for the trustees given the company's present situation.

- i) The trustees are considering various options of discontinuance. List six options available to the trustees if the scheme is discontinued. (3)

The trustees are considering offering the option of the transfer of funds to the beneficiaries to extinguish the liabilities.

- ii) Outline the issues the trustees should consider before implementing this. (4)

The trustees have decided to go ahead with a transfer of funds to the beneficiaries.

- iii)** Outline the issues the trustees need to consider when setting the terms for the transfer payments. (8)
[15]

Q. 6) A company sponsors a final salary pension scheme. The assets of the scheme are invested without any formal regard to matching the assets to the liabilities. The assets are invested in a range of equities, fixed interest and property investments.

- i)** Discuss the issues that the scheme managers would need to consider if they wished to match the value and timing of the scheme's asset proceeds with the value and timing of its liability outgo. (4)
- ii)** Outline the difficulties the scheme's managers might experience in trying to implement the strategy in part (i). (4)

For the past few years the scheme's investment objective has been to achieve an investment return in any given calendar year that is in the top quartile of returns earned by comparable benefit schemes. The scheme's investment managers have met this objective in four out of the past five years. There are no restrictions on the investment manager's choice of investments.

- iii)** Explain why this investment objective might not be appropriate. (3)
- iv)** Discuss changes that could be made to the scheme's investment objective in order to address the shortcomings set out in part (iii). (3)
[14]
